

AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH (AIUB)

Faculty of Science & Information Technology
Department of Computer Science & Engineering

PROJECT PROPOSAL

Course: Human Computer Interaction (HCI)

Recyclable Waste Detection Using Edge Device

Course	Human Computer Interaction (HCI)
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1. Overview

The Edge-Based Waste Sorting Assistant is an intelligent, on-device system that integrates lightweight computer vision models into physical disposal stations to assist users in categorizing recyclable materials efficiently and accurately. Rather than requiring users to manually identify and separate complex waste types, the system analyzes real-time camera feeds and generates immediate sorting instructions.

The interface is designed as a highly responsive, user-facing aid that provides clear, visual guidance (such as displaying “Use BLUE bin” on an attached screen) while maintaining high accuracy and low latency. Operating entirely on low-power embedded hardware, the system improves both recycling compliance and user confidence while reducing the confusion often associated with identifying diverse or overlapping materials.

The system is intended for public and commercial environments such as university campuses, corporate offices, transit stations, shopping centers, and municipal smart city infrastructure.

2. Related Work

2.1 Real-Time Object Detection Models for Waste Sorting

Recent comprehensive surveys, such as the review by Abdu and Halim (2022), highlight the growing dominance of deep learning—particularly single-stage detectors like YOLO—in waste recognition tasks. While traditional YOLO architectures achieve 70–90% mean Average Precision (mAP), they often struggle with small objects and cluttered backgrounds [1]. To address these inefficiencies, Song et al. (2025) proposed YOLO-VG, a lightweight variant of YOLOv8s tailored for embedded recyclable detection. By utilizing GSConv and ODConv convolutions alongside VoV-Ghost blocks, YOLO-VG reduced model size by 20.4% while maintaining an 88.4% mAP@0.5 [2]. Similarly, Jiang et al. (2022) developed YOLOv5s-CSS to handle cluttered rural domestic garbage, integrating an attention mechanism and a small-object detection layer to achieve 96.4% accuracy at processing speeds of ~47 frames per second [3].

2.2 Hybrid and Two-Stage Classification Architectures

To improve fine-grained material classification beyond broad bounding-box detection, recent research has shifted toward hybrid and two-stage pipelines. Umar et al. (2025) demonstrated that pairing YOLOv8 for spatial localization with a secondary Convolutional Neural Network (CNN) for classification yielded an F1-score of 0.86, outperforming standalone models [4]. Abo-Zahhad et al. (2025) applied a similar two-stage methodology for municipal bins, utilizing YOLOv5 to detect the bin and external litter, followed by YOLOv8 to classify internal waste as wet or dry [5]. Emphasizing lightweight design, Naznine et al. (2025) introduced an interpretable framework (PLDs-CNN-ridge-ELM) that cascades a 9-layer CNN with a Ridge-ELM classifier. Despite containing only 1.09 million parameters, the model achieved up to 99% accuracy in broad categorization and 96% in 12-class sub-categorization, utilizing SHAP for model interpretability [6].

2.3 Edge Computing and On-Device Inference

Deploying computer vision on resource-constrained embedded hardware remains a significant challenge. Early prototypes by Myers and Secco (2021) utilizing standard TensorFlow CNNs on a Raspberry Pi highlighted severe performance degradation, with offline accuracy dropping from 90% to 70% on-device, and inference speeds plummeting to ~1.4 FPS [7]. To improve embedded throughput, Tran et al. (2023) implemented a ResNet-50 model on a Raspberry Pi 4 to drive a physical servo-sorting bin, achieving 96% accuracy on binary (organic vs. inorganic) classification, though relying on a computationally heavy architecture [8]. More recently, Jadhav et al. (2025) evaluated YOLOv5, YOLOv7, and YOLOv8 directly on Raspberry Pi hardware for automated sorting. Their results established YOLOv8 as the optimal balance for edge devices, yielding an 83.7% mAP while maintaining real-time capabilities to control physical sorting mechanisms [9].

2.4 Datasets and Benchmarking in Waste Classification

The effectiveness of any waste classification model is heavily dependent on the quality and diversity of its training data. While datasets like TrashNet and TACO are widely used, they often lack the scale required for robust real-world generalization. Majchrowska et al. (2022) addressed this by introducing two unified benchmark datasets ("detect-waste" and "classify-waste") aggregated from multiple environmental sources. Utilizing a semi-supervised pipeline with EfficientDet-D2 and EfficientNet-B2, they achieved ~70%

Average Precision (AP) in detection and ~75% accuracy in classification, setting a baseline for multi-material environmental waste models [10].

Summary: Existing research demonstrates significant progress in waste detection using optimized YOLO variants, hybrid classifiers, and unified datasets. However, most existing solutions either rely on computationally heavy models that suffer from severe latency on embedded devices, or they sacrifice material categorization depth for speed. Furthermore, current physical sorting prototypes primarily actuate internal mechanical bins rather than guiding human behavior. The proposed Edge-Based Waste Sorting Assistant addresses these limitations by integrating highly optimized, sub-50ms lightweight models (such as MobileNet or Tiny-YOLO) onto low-power hardware (e.g., Raspberry Pi or Coral Edge TPU). By combining contextual visual analysis with immediate, user-facing screen feedback, the system bridges the gap between high-accuracy edge AI and interactive human compliance.

3. Research Gap

Despite significant advances in computer-vision methods for waste detection, a critical gap remains in the literature and in commercially available smart-recycling devices:

- Existing deep learning models for object detection are typically designed for high-end server GPUs and are too computationally heavy to run on resource-constrained embedded boards .
- Most current systems fail to achieve the real-time performance required to process images and provide immediate sorting guidance without frustrating delays at the point of disposal.
- Current models often lack robustness against real-world physical variability, struggling with cluttered backgrounds, varying illumination, occlusions, and overlapping objects common in public bins.
- Many classification approaches focus on broad binary categories rather than distinguishing the fine-grained recyclable categories necessary for accurate, multi-bin sorting.
- The integration of high-accuracy waste recognition with an intuitive, user-facing physical interfaceTherefore, there is a clear need for an edge-based waste sorting system that combines highly optimized lightweight modeling, real-time embedded inference, and clear human-interface guidance, precisely the contribution this project aims to deliver.

4. Core Principles

The proposed interface follows five established Human-Computer Interaction (HCI) principles:

4.1 Human-AI Collaboration

Rather than making recycling decisions automatically, the AI works alongside users by identifying the waste item and recommending the appropriate disposal bin. The final decision always remains with the user.

4.2 Context-Aware Waste Recognition

The AI considers both the visual characteristics of the detected object and contextual information, such as object material, shape, logos, labels, and previously recognized waste categories, to improve classification accuracy. This enables more reliable recommendations for real-world waste items.

4.3 Progressive User Guidance

Instead of overwhelming users with multiple instructions, the interface provides guidance only after a waste item has been detected. Visual indicators and concise messages appear when needed, allowing users to focus on the disposal task without unnecessary distractions.

4.4 Explainable AI Recommendations

To increase user trust and encourage learning, the interface briefly explains why a particular recycling recommendation was generated. The explanation includes the detected object, confidence score, and key visual features that influenced the prediction.

4.5 User Control and Feedback

Users maintain complete control over the final disposal decision. If the AI produces an incorrect prediction, users can report the mistake using a simple feedback option. This feedback can be incorporated into future model improvements while ensuring that users never feel forced to follow AI recommendations.

5. Example Workflow

1. The user approaches the smart recycling station carrying a waste item.
2. The user places the waste item in front of the camera.
3. The computer vision model extracts image embeddings and identifies the object.
4. The AI predicts the waste category and determines the appropriate disposal bin.
5. The display screen highlights the recommended recycling bin using text, icons, and color indicators.
6. The user disposes of the waste into the suggested bin.
7. If necessary, the user can report an incorrect prediction, allowing the system to improve future recommendations.

6. User Interface Components

6.1 Camera Capture Area

A live camera feed that continuously monitors incoming waste items and automatically captures an image when an object is detected.

6.2 Waste Detection Panel

Displays the detected waste item along with its predicted category (e.g., Plastic Bottle, Aluminum Can, Paper Cup, Glass Bottle) and the model's confidence score.

6.3 Recycling Guidance Panel

Highlights the appropriate disposal bin using clear text, icons, and color-coded indicators. Additional information such as material type and recyclability may also be displayed.

6.4 Explanation and Feedback Panel

Provides a brief explanation of the AI's recommendation, including confidence score and detected visual characteristics. Users can also select Correct Prediction or Incorrect Prediction to provide feedback, enabling continuous improvement of the AI system.

7. Research Questions

RQ1: Does the embedding-based waste detection interface improve waste classification accuracy compared to conventional image classification methods?

RQ2: Does real-time visual guidance reduce users' decision-making time when disposing of recyclable waste?

RQ3: How does providing explainable AI recommendations affect users' trust in the recycling assistant?

RQ4: Does the proposed interface improve users' recycling behavior and reduce incorrect waste disposal?

RQ5: How do users perceive the usability, usefulness, and learnability of the AI-assisted recycling system?

8. Research Hypotheses

Based on prior literature and the theoretical framework of this project, the following hypotheses are proposed:

H1: The embedding-based computer vision system will achieve significantly higher waste classification accuracy than conventional CNN-based classifiers.

H2: The AI-assisted recycling interface will significantly reduce the time required for users to identify the correct disposal bin.

H3: Explainable AI recommendations will significantly increase users' trust in the waste classification system.

H4: Users interacting with the proposed interface will report significantly higher **System Usability Scale (SUS)** scores than users of conventional recycling stations.

H5: The proposed intelligent recycling assistant will significantly reduce the percentage of incorrectly disposed recyclable waste compared to traditional recycling bins without AI assistance.

9. Evaluation Plan

9.1 Participants

Thirty to forty participants will be recruited from diverse backgrounds, including university students, office employees, and members of the general public. Participants will have varying levels of environmental awareness and familiarity with recycling practices to ensure realistic evaluation conditions.

9.2 Experimental Design

A **Within-Subjects Design** will be employed, where each participant performs recycling tasks under both experimental conditions.

Condition A (Baseline):

Participants dispose of waste using conventional recycling bins with standard labels only.

Condition B (Proposed System):

Participants use the AI-assisted smart recycling station equipped with a camera, embedding-based waste recognition, and a display that recommends the correct recycling bin.

The order of conditions will be counterbalanced to minimize learning and ordering effects. This design enables direct comparison while reducing participant variability and increasing statistical power.

9.3 Task Scenarios

Participants will complete a series of realistic waste disposal tasks involving common household and public waste items.

Examples include:

- Plastic Bottle
- Glass Bottle
- Aluminum Can
- Paper Cup
- Cardboard Box
- Plastic Food Container
- Banana Peel
- Coffee Cup
- Battery (E-Waste)
- Milk Carton

For each item, participants will identify and dispose of the waste using both the traditional recycling setup and the proposed AI-assisted system.

9.5 Statistical Analysis

The following statistical techniques will be used to evaluate the proposed system:

Paired t-test: To compare waste disposal time, classification accuracy, and incorrect disposal rates between the traditional recycling method and the AI-assisted system.

Wilcoxon Signed-Rank Test: Used when the collected data do not satisfy normality assumptions, particularly for questionnaire responses and ordinal-scale measurements.

Descriptive Statistics: Mean, standard deviation, median, and frequency distributions will summarize both objective and subjective evaluation metrics.

Effect Size (Cohen's d): To measure the practical significance of observed improvements beyond statistical significance.

Correlation Analysis: To investigate relationships among user trust, perceived usability, and recycling performance.

10. Expected Research Contributions

The proposed research is expected to make the following contributions:

- Improve recyclable waste classification through an embedding-based computer vision approach capable of recognizing diverse waste objects.
- Reduce incorrect waste disposal by providing real-time visual guidance for selecting the appropriate recycling bin.
- Minimize user decision-making time during waste disposal through intelligent AI assistance.
- Increase user trust by incorporating explainable AI recommendations with confidence scores and classification reasoning.
- Improve overall usability and accessibility of smart recycling systems for users with varying levels of recycling knowledge.

- Introduce an interactive Human-AI collaboration framework that combines computer vision, semantic embeddings, and user feedback for sustainable waste management.
- Demonstrate the effectiveness of embedding-based retrieval for handling visually similar and previously unseen recyclable objects.
- Provide empirical evidence comparing traditional recycling stations with AI-assisted recycling interfaces using a within-subjects experimental design.
- Develop HCI design guidelines for future AI-powered smart waste management systems and sustainable public interfaces.
- Contribute toward promoting environmentally responsible behavior through intelligent, explainable, and user-centered recycling technologies.